

Lumielle Academic Assistant

LIGHTWEIGHT · LOCAL · ONE-CLICK

Academic Paper Writing Assistant

From Research Topic to Final Draft — One-Stop Experience

1. Introduction & Why We Built It

Web-based AI chat is powerful, but writing a long paper comes with a few unavoidable pain points:

- **Long-conversation memory loss:** the longer the chat, the more the AI "forgets" earlier context — or even invents it.
- **Hard to revise repeatedly:** if a result isn't satisfactory, you have to regenerate it from scratch instead of editing the draft directly.
- **Context bloat:** every revision sent back into the chat makes the context longer, messier and more expensive.

Lumielle Academic Assistant turns the whole writing pipeline — **Topic Setup** → **Literature Management** → **Logic Outline** → **Chapter Writing** → **AIGC Detection & De-smelling** → **Final Word Export** — into a local tool. The AI focuses on writing one chapter at a time, while the tool manages the connections between chapters automatically, giving the model exactly as much context as it needs — never unlimited.

Design Highlights

- **Lightweight:** runs locally, no server needed, one-click deployment.
- **Local & private:** all data stays on your machine, never uploaded.
- **Smart context management:** a "context sandwich" that always gives the model the right amount of context.

2. Installation & Deployment

1 Get the package

MacOS / Windows versions. Unzip, then double-click to launch — dependencies install automatically.

2 Apply for an API Key

Apply at the DeepSeek open platform (or any OpenAI-compatible service).

3 Start using it

Double-click **Launch.command**, wait for the first-run setup, then enter your API Key in the app.

Package Layout

- **Launch.command** — one-click launcher (double-click)
- **app.py** — main program
- **.uv/** — uv runtime (auto-prepared)
- **.python/** — bundled Python 3.12 runtime
- **requirements.txt** — dependency list
- **data/** — data directory (auto-created; delete to reset)
- **models/** — optional local AIGC model (LLM engine used by default)

💡 **First launch:** the tool creates a virtual environment and installs dependencies (about 1-3 min, needs internet). After that, it starts in seconds.

3. Who It's For / Not For

✓ Great for

- Academic writing newcomers who need to quickly build a paper framework and generate a first draft.
- Undergraduate thesis and course paper writers.
- People tired of copy-pasting between AI chat windows.
- Those who want a lightweight, local and private writing environment.

✗ Not ideal for

- Deep scientific research that requires complex experiments, extensive literature databases and strict data validation.
- Those expecting fully automated, submission-ready papers without any manual revision.

4. Quick Start (5 Minutes)

- 1 Set the topic** — fill in the research topic and target word count in the sidebar. This is the global foundation; do it first.
- 2 Configure LLM** — enter your DeepSeek API Key → fetch models → pick a model.
- 3 Build the research base** — fill in research background and topic framework.
- 4 Generate** — import literature → generate outline → write chapters → export Word.

5. Recommended First-Time Path

Follow this order once to learn each module, then combine them freely to suit your workflow.

1 Sidebar global settings — set topic & length (do this first!)

Fill in the research topic and target total word count in the left sidebar.

⚠ This is the foundation of every module — the research base, logic outline and chapter writing all build on these two settings.

2 LLM Configuration — connect the "brain" first

Enter your DeepSeek API Key, fetch the model list, pick a model.

⚠ Enter the key before fetching models, otherwise it will error.

3 Research Base — build the framework

Fill in research background and expected outputs; split the research into modules. This is global context — fill it carefully, everything builds on it.

4 Literature — import your ammunition

Drag in whole folders or single PDFs; the tool parses, rates and classifies them.

5 Logic Outline → Chapter Writing → De-smell → Export

Generate the outline, write chapters, run AIGC de-smelling, and export the final Word document.

6. Module Guide

6.0 Sidebar Global Settings (do first)

The first thing after launching: fill in the **Research Topic** and **Target Total Words** in the left sidebar. This is the global foundation — the research base, logic outline and chapter writing all generate content based on these two settings.

6.1 LLM Configuration

- Enter your DeepSeek API Key (Base URL is pre-filled with the official endpoint).
- Fetch the model list, then pick a model — the flagship reasoning model is recommended (strong inference, good de-smelling results).
- Save the configuration and you're ready.

6.2 Research Base

The starting point of your research. Set up the overall framework here:

- **Research background:** topic, background materials, expected outputs.
- **Module management:** split the research into modules and fill in content requirements one by one.

All settings here are synced to the logic outline and chapter writing as global context.

6. Module Guide (continued)

6.3 Literature Processing

- **Batch upload:** drag in whole folders or single PDFs.
- **Auto parsing:** extracts title, abstract, authors and more.
- **Rating & classification:** each paper is rated and categorized by the LLM; chapter writing only cites papers that passed the rating.

6.4 Logic Outline (core module)

The logical skeleton of your paper is built here, with multi-level outline support:

- **Outline generation & editing:** auto-generate a multi-level outline; every node can independently edit title, description and word count; add/delete nodes freely.
- **Node editing:** parent nodes act as chapter containers; leaf nodes are the actual content entities.
- **Chart instructions:** each leaf node can be configured with chart-drawing instructions; the LLM auto-generates charts and mounts them to the chapter.
- **Personal experiment marking:** chapters marked as personal experiments are treated as original research / first-hand data when generating text.
- **Word count management:** per-node word allocation at a glance, with whole-tree statistics to keep length under control.

6. Module Guide (continued)

6.5 Chapter Writing

Single-chapter writing & editing:

- **Chapter navigation:** written chapters show a  mark; parent nodes show subtree progress at a glance.
- **Global context:** before generating, the topic / outline / upstream memory / downstream boundary / bound references are all displayed.
- **AI generation:** auto-corrects word count; compresses memory after generation for later chapters to stay consistent.
- **Charts & images:** charts are auto-generated and mounted; images can be uploaded with explanations.

Batch workbench:

- Tree-style collapsible navigation; parent nodes are containers, only leaf nodes can be queued.
- Two independent queues: chapter generation / chart drawing, with one-click select-all.
- Batch-generate chapters with per-chapter word-count correction and a live progress bar.

6.6 AIGC Detection & De-smelling

- **Detect:** pick a chapter and click Detect to see the AI probability percentage. Two engines available (LLM / local model).
- **De-smell:** select chapters with high AI probability and click De-smell; the tool polishes them repeatedly until the AI flavor drops, then shows the result. A manual read-through after de-smelling is recommended.

6.7 Prompt Configuration

The global "flavor adjustment zone" controlling the paper's style and de-smelling strength. Defaults are tuned for academic scenarios — newcomers can leave them as-is.

7. FAQ

Q: Where is my API Key stored?

A: Only in the local configuration file — never uploaded to any server.

Q: Will my data be lost?

A: Topic, literature and drafts are all persisted in the local **data/** directory. Nothing is lost on normal exit.

Q: Are the detection results reliable?

A: This is a lightweight assistant for reference only. For important occasions, cross-validate with professional detection platforms.

Q: Do I need an internet connection?

A: Only for the first-run dependency install and LLM API calls. The app itself runs fully locally.

Q: Can I reset all data?

A: Delete the **data/** folder and relaunch.

8. Data Safety

Localization = privacy. Your paper content belongs only to you — it stays in the local **data/** directory and is never uploaded to any server. The tool runs entirely on your machine.

9. License

This tool is provided under the MIT License for personal learning use. Resale for profit is prohibited.



Finally, the developer insists on showing off this beautiful poster.

— No wait, it's *me* who insists. After all, who could resist me looking this good? ✧